

Gene Pulser® Electroporation

Cell Type	Bacterial, gram positive	Molecules Electroporated	DNA: plasmid, 6 to 12 kB.
Species Used	<i>Streptococcus sanguis</i> , FW213		

Before the Pulse

Cell Growth Medium	Brain Heart Infusion (BHI) +0.2% glucose	Growth Phase at Harvest	O.D. (600) = early log phase
		Pre-pulse Incubation	10 mM Tris-Cl (pH 4.0), 0.5 M sucrose
Wash Solution	10 mM Tris-Cl (pH 6.0), 0.5 M sucrose		

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	4 °C		
Electroporation Medium	10 mM Tris-Cl (pH 4.0), 0.5 M sucrose	Cuvette Gap	0.2 cm
Cell Density	10 (9) cells / ml	Voltage	2.5 kV
Volume of Cells	50 µl	Field Strength	12.5 kV/cm
DNA Concentration	50 µg / ml		
DNA Resuspension Buffer	1x TE buffer (10 mM Tris, 1 mM EDTA, pH 8.0)	Capacitor	25 µF
Volume of DNA	1 to 2 µl	Resistor	(Pulse Controller) 200 Ω
After the Pulse		Time Constant	4.0 to 4.3 msec
Outgrowth Medium	THB (Difco) + 0.625 M sucrose		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	2 to 3 hours
Selection Method or Assay Used	antibiotic selection
Electroporation Efficiency	10 (4) transfectants / µg, using supercoiled plasmid DNA
Per Cent Survival	30 to 40 %

Name of Submitter Chris Fenno

Institution Address University of Vermont
Dept. Microbiology & Molecular Genetics
B227 Given
Burlington, VT 05405

Survey Number
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